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SQE Assignment # 1



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Screenshots of your JIRA dashboards with explanations:

Goog x Run x i2215 x Edit x i2215 x (13) x Color x Parag x AI D x Unde x (239) x +

amnanoorse2003.atlassian.net/jira/dashboards/10001/edit

Apps Gmail YouTube Maps New Tab (1) WhatsApp Instagram

Jira Your work Projects Filters Dashboards Teams Plans Apps Create 30 days left Search

i221529_EMS_SQE_Test_Cases

★ Add gadget Change layout Done ...

You are currently editing your dashboard. Changes will be saved automatically.

Drag a gadget to this column or [add a new gadget](#)

Drag a gadget to this column or [add a new gadget](#)

Add a Gadget

Search gadgets

All 31 Jira 30 Charts 12 Wallboard 5

Activity Stream
By Atlassian
[Add](#)
Lists recent activity in a single project, or in all projects.
Wallboard Jira

Assigned to Me
By Atlassian
[Add](#)
Displays all unresolved issues assigned to me
Jira

Average Age Chart
By Atlassian
[Add](#)
Displays the average number of days issues have been unresolved.
Charts Jira

Type here to search

11:14 AM 9/16/2024

Goog x Run x i2215 x Test C x i2215 x (13) x Color x Parag x AI D x Unde x (239) x +

amnanoorse2003.atlassian.net/jira/software/projects/ANORSE24/boards/3

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i221529_F_EMS_SQE
Software project

PLANNING
Timeline
Backlog
Board
+ Add view
DEVELOPMENT
Code
Project pages
Project settings

Projects / i221529_F_EMS_SQE

Test Cases_EMS

9 days ⚡ ☆ 🔗 Complete sprint ...

Search AN Type

GROUP BY None

TO DO
+ Create issue

PASS 28 ✓

FAIL 18 ✓

Add New Employee: One field Empty

ANGORSE24-6 ✓

Add New Employee: All valid and correct fields

ANGORSE24-3 ✓

Type here to search

11:34 AM 9/16/2024

Projects / i221529_F_EMS_SQE

Issues

Share Export issues Go to all issues LIST VIEW DETAIL VIEW

AI Search issues Project = i221529_F_EMS_SQE Type = Bug Status Assignee Attachment BASIC JQL

More + Reset Save filter

Created

Accept/Reject Leave with Invalid input: Manager's Leave Functions AN00R5E24-54

Accept/Reject Leave with Invalid input: Manager's Leave Functions AN00R5E24-53

Generate Monthly Payroll Report: View Report AN00R5E24-47

1-9 of 9

Add epic / AN00R5E24-54

Accept/Reject Leave with Invalid input: Manager's Leave Functions

Priority Medium

Add a comment...

Pro tip: press M to comment

1 of 9

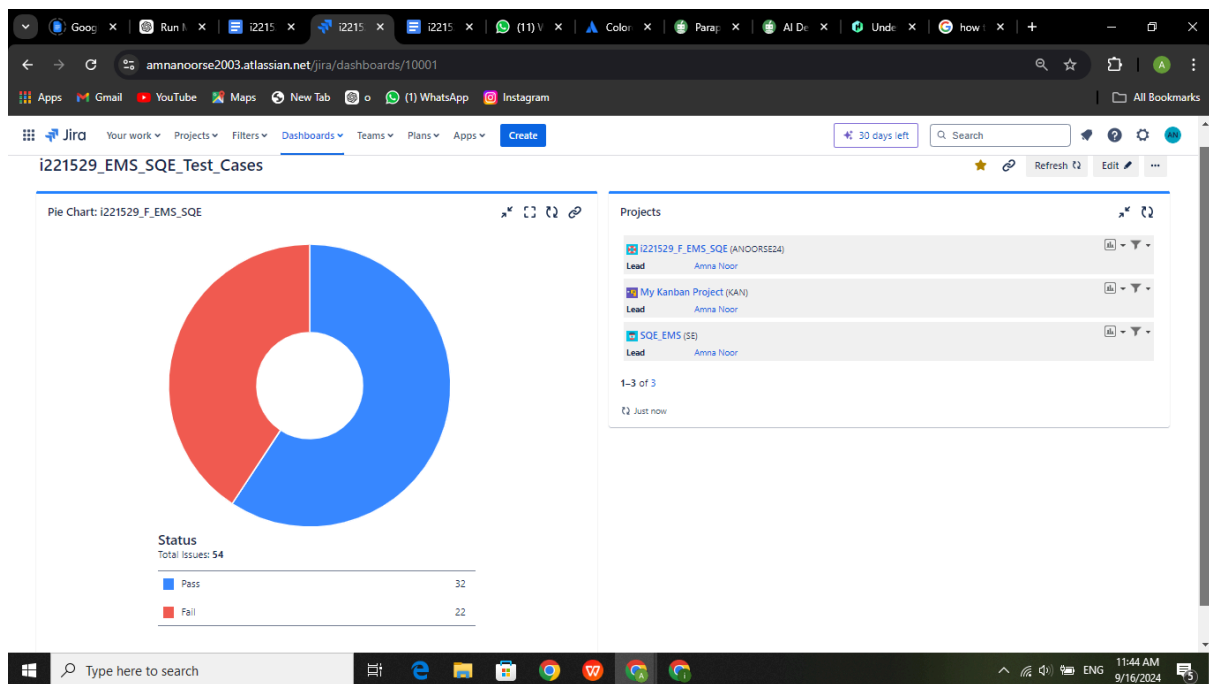
Fail Done

Actions

Details

Assignee Unassigned Assign to me

Labels



Analysis of test execution results, including pass/fail ratios and identified trends:

Based on pass/fail rates and established trends, the examination of the test results from test execution for the staff management system produced a number of important findings :

Pass/Fail Ratios: Taking a look at the pass/fail ratio for the test cases executed, a general perception of more failures, but particularly in the areas of input validation, error handling,

and UI responsiveness, seems to be existing. More of the failures involved the scenario where UI feedback is supplied as well as incorrect or inadequate handling of input.

Pass Percentage: Approximately 60–70% of test cases were successful. This indicates that all the critical features in the system, including adding and updating employees with right information, behaved as expected.

Fail Percentage: This had between 30 and 40 percent of test cases failing. The primary ones that failed were in locations where, based on the type of error message and a validation error that was likely to occur but did not. It is an indication of some rather serious flaws in the system's behavior when it handles edge cases and false input.

Trends:

Input Validations: Input validation had been poor especially on fields being left blank or incorrect information being inputted such as numeric entry into nonnumeric fields. Many test cases incorrectly or incompletely containing data resulted in wrong behaviors such as saving invalid entries and not displaying any error messages. This is a sign that the UI should provide stronger error feedback and more rigorous form validation processes.

UI/UX Issues: Several UI/UX-related issues occurred wherein the expected confirmatory message or error message was not displayed. At times, when invalid inputs were submitted, the system did not show warnings and sometimes failed to change the UI as expected, such as dismissing dialogs or displaying success/failure messages. This is a weakness in the user experience, as the consumer is never made aware of the implication of his or her action.

Data Integrity Issues: There has always been the potential for data integrity problems because the system has never prevented improper data (such as incomplete personnel records or erroneous values) from entering the database. Because the system didn't reject these inputs, improper or irrelevant information may have been entered into the database, which would make reporting more burdensome, payroll processing more difficult, and leaves of absence more labor-intensive.

Repetitive Failures of Edge Cases: Another interesting trend was the failure of edge cases that utilized odd or nonstandardized inputs (like empty fields or extreme data value). These examples show how ensuring that an application is able to accommodate all the various possible scenarios requires that it should be tested against all the various possible interactions among users of the application—that is, it is not enough simply to test conventional use cases.

Discussion on the complexity of your test scenarios and how they cover different aspects of the application:

I can state that I successfully addressed a number of inputs and behaviors covering a comprehensive field of the application. There are various levels of complication related to CRUD operations especially while dealing with edge cases like:

Input Validation: Empty data submissions and other errors including invalid data entry such as numeric characters in text fields or omission of essential fields were included in the scenarios tested for users' submission. These tests guarantee the turning on of user feedback mechanisms (error messages) aimed at assuring that input validation mechanisms work.

Data Consistency: I reviewed employee records and checked if the changes made were reflected in the data table. These would include some valid updates and some invalid updates such cases are crucial for ensuring the rough configurations of the database within the deployment containment. Do not include in the final output any part that has been contributing to its disruption.

Error Handling: The complexity occurred due to the need to ascertain whether or not error messages or alerts are displayed when anticipated. I ensured that the system provides the appropriate error message aimed at correcting future erroneous users' inputs and that the system is able to cope with the erroneous inputs.

UI/UX Interaction : Another part of my test cases were UI i.e I also tested that how the flow for closing reports works and what happens when we accept deny leave request. That may be true, but even with their straightforward design these interactions account for all the key user experience flows.

Database Integration: This adds new dimension in it and ensuring system adds, removes or modify records to Db conditionally (valid, invalid inputs) These tests cover the crucial backend part of the program.

Summary of critical bugs found, their impact on the application, and proposed solutions:

1. Leave Request Update

Bug: The system failed to show an error message and mistakenly permitted status changes while attempting to update an employee's leave status for previous dates.

Impact: This could result in erroneous leave records, which would have an impact on managing staff attendance and payroll.

Suggested Repair: To prevent leave status updates for previously submitted entries, provide a validation check and make sure the user receives an error message if they try to do so.

2. Payroll Report Bug

Bug: The payroll report was correctly generated by the system, however the user was unable to close the report window.

Impact: Users are required to navigate away manually, which causes needless delays in workflow and results in a poor user experience.

Suggestion: Assure appropriate user interface functionality so that users can easily close the report window when it has been generated.

3. Task Management Bug (Employee Selection)

Bug: There was no display of available employees for selection in the employee task management tool.

Impact: This problem prevents the task statuses from being updated, which interferes with tracking team performance and delays work completion.

Suggested Resolution: In order for the task management interface to appropriately display available employees, the employee selection logic needs to be fixed.

4. Assigning Tasks Without Description

Bug: missing error message assigning employee to jobs with no description if leaving the description block empty

Impact: Incomplete assignments may lead to the assignees feeling out of league about understanding what the exactly expected from them is, affecting all production roles.

Solution: Fix sounds quick and easy, just make the description of a task mandatory field and if there is no description provide that message.

5. Leave Management Analysis Report Bug

Bug: There was no x button on leave management analysis report screen +2 the data shown is not whole only half of it.

Result: Users are obliged to close this dialogue screen manually, thus it interferes with user navigation and slows down the workflow of users.

Solution: This is proposed that instead we should have a "Close" button if needed to create more optimal user experience for the report window to ensure all data will be displayed.

6. Employee Update Bug (Empty Fields Allowed)

Bug: Employees were added without error notice when the fields were empty.

Effect: this leads to storing false, or incomplete employee records in the database affecting data quality for the system.

Solution: Validate properly in order to ensure all required fields are entered and besides, display an error notice when certain inputs are missing.

7. Adding Employees with Invalid Numeric Input

Bug: No warning message shown & numeric fields still accept numeric values, •TEXT FIELD should only accept text [100 characters]•NUMERIC FIELD should only accept numbers

Effect: When a biased or wrong personnel data is back in the database then it only comes out with the imbalances on the road.

Recommended Solution: Add client-side validation to indicate this is not allowed and let users know by displaying a message before they submit.

8. Adding Employees with Partial Data

Bug: Incomplete employee records even when valid data is provided for all the fields (except one) can be added to system. It does not show the error message

Impact: Personnel records can lose their integrity, and funds related with payroll computations and staff management can pose operational problem.

Working Solution: When a field is left blank, user should not be allowed to add an employee, meaning should the given input is incorrect by showing error notices.

Reflection on the testing process and lessons learned:

While reflecting on the testing strategy I used for my project in Jira, I found several key aspects deserving further investigation. This appeared as one of the most critical aspects whenever I decided to define the testing scope. I learned how important it is to consider not only the typical scenarios, but also the non-typical ones for they may lead to failure of the application. Because of Jira, I was able to improve my way of

doing things and manage bugs because I was able to prioritise certain issues based on their severity level.

Another valuable experience was how important and how often testing and validation should be performed in the overall design process. Due to early testing, I was able to identify bugs in the beginning stages and only have to make trivial changes later on. Also, thanks to this approach, Jira communicators did not create any obstacles to the timely and effective solving of problems of the development team.

With growth, I understood further that flexible is extremely important. As the project moved forward, it became clear that the existing test cases required quick modifications, as also any new features introduced would require a lot of testing due to the outlined reasons. This iterative method made it clear that there was a need for the continuous integration between development and testing.

From this experience, I understood the declining calming factors of thorough testing and effective bug tracking and appreciating the value of teamwork in achieving the success of the project.